SUDARSHAN ANAND

Atlanta, GA (willing to relocate)|asudarshan14@gmail.com|+1 (404) 453-5664|LinkedIn: www.linkedin.com/in/sanand315/
Portfolio: sudarshananand14.vercel.app | **F-1 OPT (2026–2029)**

EDUCATION

Georgia Institute of Technology — Atlanta, GA, USA
Master of Science in Computational Science and Engineering (CGPA: 4.0/4.0) — May 2026

Birla Institute of Technology and Science (BITS Pilani) — Pilani, Rajasthan, India
Master of Science in Mathematics and Bachelor of Engineering in Computer Science (CGPA: 9.4/10) — Aug 2024

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, PostgreSQL, Neo4j, Git, Linux/Unix
Machine Learning & AI: PyTorch, multimodal learning, computer vision, self-supervised learning, LLM applications, retrieval-augmented generation (RAG), prompt engineering, time-series foundation models, transformer architectures, model fine-tuning, model evaluation, inference optimization, feature engineering, A/B testing, statistical modeling
Systems & Infrastructure: FastAPI, Pinecone, ChromaDB, GCP, SLURM, scalable ML pipelines, cloud deployment, API development, backend systems, ML workflow automation, CI/CD, production ML systems
Data Science & Analytics: Data preprocessing, exploratory data analysis (EDA), performance optimization, predictive modeling, longitudinal data analysis, unstructured data processing
Professional: Cross-functional collaboration, stakeholder-facing collaboration, technical documentation, research communication, project leadership

PROFESSIONAL EXPERIENCE

AI Product Development Intern — Jun 2025 – Aug 2025
Rezolve.ai — Dublin, CA, USA [Remote]

- Built an enterprise AI search assistant (RAG-based) that reduced manual policy lookup effort by integrating FastAPI, Pinecone, and PostgreSQL to support contextual retrieval and follow-up question generation.
- Improved search responsiveness for enterprise knowledge discovery by developing a low-latency autocomplete pipeline with personalized and AI-generated suggestions deployed on GCP Cloud Run.
- Prototyped an LLM-based incident triage assistant that converted noisy infrastructure alerts into structured response workflows for IT support operations.

AI Scientist Intern — Jan 2024 – Jun 2024
Qure.ai Technologies Pvt. Ltd. — Bangalore, Karnataka, India

- Improved clinical lung nodule prediction consistency by ~45% over prior internal baselines through PyTorch-based computer vision model optimization.
- Evaluated model stability across longitudinal imaging studies to validate deployment readiness for clinical workflows.

RESEARCH AND PROGRAMMING PROJECTS

Multimodal Progression Tracking of Neurodegenerative Disorders, *Georgia Institute of Technology* — Aug 2025 – present

- Built a multimodal Parkinson's disease progression pipeline using MRI scans and demographic data across 3,000+ patient records using distributed GPU training on SLURM clusters.
- Fine-tuned self-supervised vision models with demographic feature fusion to improve disease progression classification performance (Macro F1 ≈ 0.60).

Investigating importance of Patient Metadata in Public Health, *Edith Cowan University* — May 2025 – present

- Leading a cross-functional team as the sole data scientist in analysing 1.2M unstructured patient records to quantify the impact of extreme metadata sparsity (94% missing rate) on predictive performance and downstream decision-making.
- Evaluated Gemma-2 and MedGemma models under high missing-data conditions to study robustness and reasoning reliability in public-health prediction tasks.

Samay: Modular ML Inference Framework, *Georgia Institute of Technology* — Jan 2025 – May 2025

- Developed **Samay**, an open-source framework unifying 10 time-series foundation models, enabling standardized inference, fine-tuning, and evaluation workflows across diverse data modalities (34 stars, 7 forks).
- Built modular PyTorch 2.0+ pipelines to enable zero-shot evaluation and seamless task-specific adaptation (SFT), optimizing for fast execution across diverse data modalities.
- Architected and executed large-scale benchmarking protocols, validating model performance, scalability, and baseline metrics against unified evaluation standards.

HONORS & AWARDS

IEEE BHI 2025 Data Challenge Competition Winner — Oct 2025

- Engineered and developed an end-to-end AI/ML-based depression risk prediction system, contributing to a winning solution in the IEEE-sponsored competition (supported by NSF and Google)

IEEE BHI 2025 Young Professional NextGen Scholar (NSF, EMBS, Google Sponsored) — Sept 2025

SELECTED PUBLICATIONS

Foundation Models / ML Systems

- H. Kamarthi, S. Li, S. Anand, S. Das, A. Ramesvaran, A. Bhalerao and B.A. Prakash, *"Samay: A Unified Pipeline Framework for Time-Series Foundation Models"* VLDB 2026 Demo Track Submission